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ABSTRACT

American and Bermudan options stand out for their flexibility in early exercise, presenting unique challenges in pricing. This paper introduces an enhanced method for efficiently pricing these options using an Almost-Exact Scheme (AES) within Heston-type stochastic volatility models. A Bermudan option is an American-style financial option that permits early exercise only on pre-specified dates. The Monte-Carlo simulation, a standard approach for pricing these options, gains efficiency when paired with an exact simulation scheme by simulating only on these early exercise dates. Such an exact scheme exists for the classical Black-Scholes model since the corresponding stochastic differential equation has a closed-form solution. For the more recent Heston model with one stochastic volatility process, a so-called AES uses the non-central chi-square probability distribution for the variance process.

This paper extends this methodology to the more complex Double Heston model. It shows the analytical derivation of the AES for the Double Heston model and numerically explores its benefits for both models. We compared AES with Euler and demonstrated cases when AES works better than Euler. Findings reveal that AES demonstrates efficacy when the number of simulation steps aligns with the exercise dates, making the AES efficient for pricing American and Bermudan options.

KEYWORDS

Almost Exact Scheme; Monte Carlo; Bermudan Options; Least Squares Monte Carlo; CIR; Heston Model; Double Heston Model; Stochastic Volatility

1. Introduction

The challenge of accurately pricing American and Bermudan options, known for their early exercise flexibility, has been a significant area of research in financial mathematics. This paper introduces an improved method for efficiently pricing these options using an Almost-Exact Scheme (AES) within Heston-type stochastic volatility models. The AES aims to address the limitations of traditional numerical methods, such as the Euler-Maruyama and Milstein schemes, by providing a more precise simulation, especially for small time steps.

Added part:

[?] studied the idea of an exact simulation of the sample stock price and variance from the exact distribution and showed that their method has a faster convergence rate of the error compared to the Euler discretization method. [?] and [?] consider some approximations of the exact schemes.

The pricing of American and Bermudan options has been extensively studied using various numerical techniques. The Alternating Direction Implicit (ADI) schemes have been effective for American options under the Heston model, as detailed by [?]. Penalty methods, explored by [?], provide another approach for handling the complexities of American options with stochastic volatility. As discussed by [?], the operator splitting methods have also shown promise in addressing the challenges of stochastic volatility in American options.

Moreover, space-time adaptive finite difference methods, highlighted by [?], offer a robust framework for pricing American options by dynamically adjusting the grid to capture the option's features more accurately. The use of multigrid methods, as proposed by [?], facilitates the efficient resolution of linear complementarity problems associated with American-style options. [?] contributed to this field by demonstrating the effectiveness of multigrid solutions for two-factor American options.

For Bermudan options, the Fourier-based valuation method by [?] stands out for its application under the Heston model, providing a reliable framework for pricing these options. [?] introduced a tree-based method tailored explicitly for American options in the Heston model, further improving the computational efficiency and accuracy.

These methods have laid a solid foundation for the development of advanced techniques in option pricing. However, each comes with its own set of challenges and limitations. The ADI schemes, while effective, can be computationally intensive. Penalty methods require careful parameter tuning to ensure accuracy and stability. Although robust, operator splitting methods and space-time adaptive finite difference methods may still face difficulties in handling the path-dependent nature of early exercise options.

The AES presented in this paper seeks to overcome these challenges by using the specific properties of the Heston model and its extensions. By incorporating an exact simulation of the variance process, the AES significantly improves precision over traditional numerical methods when the number of simulation steps is small. This approach ensures that the variance process remains positive as the simulation is done directly using the exact distribution, addressing one of the key issues in simulating stochastic volatility models.

This study discusses the AES for both the Heston and Double Heston models to evaluate its performance in pricing American and Bermudan options. The following sections consider the AES's theoretical framework, implementation, and numerical experiment results.

2. AES Schemes of the Heston and Double Heston Models

The literature suggests that simulations of asset prices using an exact solution work better for smaller-time discretizations. In addition, standard schemes (Euler-Maruyama, Milstein) do not work well with positive processes such as the variance process as suggested by [?]. The reason behind the statement is that the probability mass function of the variance process is around zero, and one such example is the CIR process.

For the stated reasons, we consider the Almost Exact Scheme (AES) under the

Heston model as suggested in [?]. The scheme deals with a solution to the CIR process, and the issue of it becoming less than zero in a simulation is removed, as we will sample directly from its non-negative distribution. That's where the name comes from. The AES for the Heston model is given by

$$x_{i+1} \approx x_i + c_0 + c_1 v_i + c_2 v_{i+1} + \sqrt{c_3 \cdot v_i} Z_1, \quad (1)$$

and the CIR process [?](add ref by Cox et al. (1985)) simulated as follows

$$v_{i+1} \approx \bar{c}(t_{i+1}, t_i) \cdot \chi^2(\delta, \bar{\kappa}(t_{i+1}, t_i)), \quad (2)$$

with the following parameters:

$$\bar{c}(t_{i+1}, t_i) = \frac{\gamma^2}{4\kappa} \left(1 - e^{-\kappa(t_{i+1}-t_i)}\right), \quad \bar{\kappa}(t_{i+1}, t_i) = \frac{4\kappa e^{-\kappa(t_{i+1}-t_i)}}{\gamma^2 (1 - e^{-\kappa(t_{i+1}-t_i)})} \cdot v_i,$$

and $\chi^2(\delta, \bar{\kappa}(\cdot, \cdot))$ represents the noncentral chi-squared distribution with δ degrees of freedom and noncentrality parameter $\bar{\kappa}(\cdot, \cdot)$. The parameters are

$$c_0 = \left(r - \frac{\rho_{1,2}}{\gamma} \kappa \bar{\nu}\right) \Delta t, \quad c_1 = \left(\frac{\rho_{1,2}}{\gamma} \kappa - \frac{1}{2}\right) \Delta t - \frac{\rho_{1,2}}{\gamma}, \quad c_2 = \frac{\rho_{1,2}}{\gamma}, \quad c_3 = (1 - \rho_{1,2}^2) \Delta t.$$

The idea presented in [?] will be extended to the Double Heston model with the same motivation as we presented.

Hence, now we are concentrating on obtaining an extension of the AES to the Double Heston model given by Eq. (3). By applying Itô's lemma to the log transformation $X_t := \ln S_t$ and the Cholesky decomposition of the correlation matrix, we have the following

$$\begin{aligned} dX_t &= \left(r - \frac{1}{2}(\nu_t^1 + \nu_t^2)\right) dt + \sqrt{\nu_t^1} \left(\rho_{1,3} d\widetilde{W}_t^3 + \sqrt{1 - \rho_{1,3}^2} d\widetilde{W}_t^1\right) \\ &\quad + \sqrt{\nu_t^2} \left(\rho_{2,4} d\widetilde{W}_t^4 + \sqrt{1 - \rho_{2,4}^2} d\widetilde{W}_t^2\right), \\ d\nu_t^1 &= \kappa_1 (\bar{\nu}_1 - \nu_t^1) dt + \gamma_{v_1} \sqrt{\nu_t^1} d\widetilde{W}_t^3, \\ d\nu_t^2 &= \kappa_2 (\bar{\nu}_2 - \nu_t^2) dt + \gamma_{v_2} \sqrt{\nu_t^2} d\widetilde{W}_t^4, \end{aligned} \quad (3)$$

where $d\widetilde{W}_t^1, d\widetilde{W}_t^2, d\widetilde{W}_t^3$ and $d\widetilde{W}_t^4$ are independent Brownian increments.

By doing this, we get the system of three SDEs in which each variance process is driven independently by Brownian motions and where $X(t)$ is correlated to ν_t^i for $i = 1, 2$. Therefore, we can use the properties of the marginal distribution of ν_t^i .

Taking integration of all processes in a certain time interval $[t_i, t_{i+1}]$, we get the

following discretization scheme

$$\begin{aligned}
x_{i+1} &= x_i + \int_{t_i}^{t_{i+1}} \left(r - \frac{1}{2} (\nu_t^1 + \nu_t^2) \right) dt + \rho_{1,3} \int_{t_i}^{t_{i+1}} \sqrt{\nu_t^1} d\widetilde{W}_t^3 \\
&\quad + \sqrt{1 - \rho_{1,3}^2} \int_{t_i}^{t_{i+1}} \sqrt{\nu_t^1} d\widetilde{W}_t^1 + \rho_{2,4} \int_{t_i}^{t_{i+1}} \sqrt{\nu_t^2} d\widetilde{W}_t^4 \\
&\quad + \sqrt{1 - \rho_{2,4}^2} \int_{t_i}^{t_{i+1}} \sqrt{\nu_t^2} d\widetilde{W}_t^2, \\
v_{i+1}^1 &= v_i^1 + \kappa_1 \int_{t_i}^{t_{i+1}} (\bar{\nu}_1 - \nu_t^1) dt + \gamma_{v_1} \int_{t_i}^{t_{i+1}} \sqrt{\nu_t^1} d\widetilde{W}_t^3, \\
v_{i+1}^2 &= v_i^2 + \kappa_2 \int_{t_i}^{t_{i+1}} (\bar{\nu}_2 - \nu_t^2) dt + \gamma_{v_2} \int_{t_i}^{t_{i+1}} \sqrt{\nu_t^2} d\widetilde{W}_t^4.
\end{aligned} \tag{4}$$

Now we can notice the same integrals that include $\widetilde{W}_t^i, i = 3, 4$ appear in the SDEs. Using this, we get that the discretization for x_{i+1} is given by

$$\begin{aligned}
x_{i+1} &= x_i + \int_{t_i}^{t_{i+1}} \left(r - \frac{1}{2} (\nu_t^1 + \nu_t^2) \right) dt + \sqrt{1 - \rho_{1,3}^2} \int_{t_i}^{t_{i+1}} \sqrt{\nu_t^1} d\widetilde{W}_t^1 \\
&\quad + \frac{\rho_{1,3}}{\gamma_{v_1}} \left(v_{i+1}^1 - v_i^1 - \kappa_1 \int_{t_i}^{t_{i+1}} (\bar{\nu}_1 - \nu_t^1) dt \right) \\
&\quad + \frac{\rho_{2,4}}{\gamma_{v_2}} \left(v_{i+1}^2 - v_i^2 - \kappa_2 \int_{t_i}^{t_{i+1}} (\bar{\nu}_2 - \nu_t^2) dt \right) \\
&\quad + \sqrt{1 - \rho_{2,4}^2} \int_{t_i}^{t_{i+1}} \sqrt{\nu_t^2} d\widetilde{W}_t^2.
\end{aligned}$$

As in the Euler scheme, the integrands are approximated by their values at the left-side boundary of the integration interval. Using this, when $\Delta t = t_{i+1} - t_i$ the discretization for x_{i+1} is given by

$$\begin{aligned}
x_{i+1} &\approx x_i + \left(r - \frac{1}{2} (v_i^1 + v_i^2) \right) \Delta t + \frac{\rho_{1,3}}{\gamma_{v_1}} (v_{i+1}^1 - v_i^1 - \kappa_1 (\bar{\nu}_1 - v_i^1) \Delta t) \\
&\quad + \sqrt{1 - \rho_{1,3}^2} \sqrt{v_i^1} (\widetilde{W}_{t_{i+1}}^1 - \widetilde{W}_{t_i}^1) + \frac{\rho_{2,4}}{\gamma_{v_2}} (v_{i+1}^2 - v_i^2 - \kappa_2 (\bar{\nu}_2 - v_i^2) \Delta t) \\
&\quad + \sqrt{1 - \rho_{2,4}^2} \sqrt{v_i^2} (\widetilde{W}_{t_{i+1}}^2 - \widetilde{W}_{t_i}^2).
\end{aligned} \tag{5}$$

Finally, using the property of normally distributed random variables and substituting into Eq. (5) we get that the AES for the Double Heston model is given by

$$x_{i+1} \approx x_i + c_0 + c_1 v_i^1 + c_2 v_i^2 + c_3 v_{i+1}^1 + c_4 v_{i+1}^2 + \sqrt{c_5 \cdot v_i^1} Z_1 + \sqrt{c_6 \cdot v_i^2} Z_2,$$

and the two CIR processes

$$v_{i+1}^1 \approx \bar{c}^1(t_{i+1}, t_i) \cdot \chi^2(\delta^1, \bar{\kappa}^1(t_{i+1}, t_i)), \tag{6}$$

$$v_{i+1}^2 \approx \bar{c}^2(t_{i+1}, t_i) \cdot \chi^2(\delta^2, \bar{\kappa}^2(t_{i+1}, t_i)), \tag{7}$$

with the following parameters:

$$\bar{c}^j(t_{i+1}, t_i) = \frac{\gamma_{V_j}^2}{4\kappa_j} \left(1 - e^{-\kappa_j(t_{i+1}-t_i)}\right), \quad \bar{\kappa}^j(t_{i+1}, t_i) = \frac{4\kappa_j e^{-\kappa_j(t_{i+1}-t_i)} \cdot v_i^j}{\gamma_{V_j}^2 (1 - e^{-\kappa_j(t_{i+1}-t_i)})},$$

for $j = 1, 2$, and $\chi^2(\delta, \bar{k}(\cdot, \cdot))$ represents the noncentral chi-squared distribution with δ degrees of freedom and noncentrality parameter $\bar{k}(\cdot, \cdot)$. The constants are

$$\begin{aligned} c_0 &= \left(r - \frac{\rho_{1,3}}{\gamma_{V_1}} \kappa_1 \bar{v}_1 - \frac{\rho_{2,4}}{\gamma_{V_2}} \kappa_2 \bar{v}_2 \right) \Delta t, \\ c_1 &= \left(\frac{\rho_{1,3}}{\gamma_{V_1}} \kappa_1 - \frac{1}{2} \right) \Delta t - \frac{\rho_{1,3}}{\gamma_{V_1}}, \\ c_2 &= \left(\frac{\rho_{2,4}}{\gamma_{V_2}} \kappa_2 - \frac{1}{2} \right) \Delta t - \frac{\rho_{2,4}}{\gamma_{V_2}}, \\ c_3 &= \frac{\rho_{1,3}}{\gamma_{V_1}}, \\ c_4 &= \frac{\rho_{2,4}}{\gamma_{V_2}}, \\ c_5 &= (1 - \rho_{1,3}^2) \Delta t, \quad c_6 = (1 - \rho_{2,4}^2) \Delta t. \end{aligned}$$

Hence, the AES for the Double Heston model is obtained.

3. Evaluating AES for American and Bermudan Option Pricing

The section's primary goal is to check the accuracy and behavior of the AES for pricing American and Bermudan options. To accomplish that, we apply multiple techniques to the Monte Carlo simulation. Finally, we provide specific comments for the Heston and Double Heston models.

Specifically, the focus is on analyzing option prices using a small number of time steps when simulating the asset price process. The idea of considering a small number of time steps comes from the fact that the time discretization error in the AES comes only from the asset price process, as the CIR process is simulated using the exact scheme, that is, the noncentral chi-square distribution, which is always positive.

The Euler-Maryama and Milstein schemes are highly dependent on a chosen method (reflection, full truncation) to ensure that the variance process stays positive. In contrast, in the AES, there are no such issues. This has been addressed in [?].

In addition, the two variance reduction techniques (see [?]) used to generate asset prices in the paper are antithetic paths and moment matching. Those methods proved to improve the accuracy of the simulation.

However, the proposed control variate method that impacts the accuracy the most out of the variance reduction methods was not used in the paper to fully capture the influence of the AES on the price. As there is an efficient COS method for pricing European options, it could be used as a control variate to improve accuracy further. The variance reduction methods do not increase computational time, which makes them applicable.

We use the [?] Least Square Monte Carlo (LSM) method to price American and Bermudan options. The conditional expectation function is estimated using standard

polynomials as basis functions of up-to-order three and in-the-money paths to get better estimates unless specified otherwise.

Even though the LSM method is robust to the choice of basis functions, we compare and comment on choices in this paper to capture their influence on the AES. In addition, as suggested in [?], when estimating the conditional expectation function, instead of the asset prices, we use normalized values obtained by dividing asset prices by the strike prices.

3.1. AES under the Heston Model

This section aims to investigate the impact of AES on the valuation of American and Bermudan options under the Heston model when the number of time steps is small. To do that, we will consider different sets of parameters for the Heston model.

Throughout the numerical analysis, the simulation will consistently have a fixed number of paths, with $N = 1,000,000$, and a total of 20 simulations will be conducted.

American put options. The first set of parameters to be examined for pricing American options, given in Eq. (8), has been widely employed in various articles, including [?], [?], [?], [?], [?], [?], and [?]. The reference price used to calculate the relative errors was obtained as an average of the American option prices authors of those articles got.

Let's first consider the following parameters, given by Eq. (8), that satisfy the Feller condition given by Eq. (??)

$$\begin{aligned} K = 10, \quad v_0 = 0.0625, \quad T = 0.25, \quad r = 0.1, \\ \gamma = 0.9, \quad \bar{\nu} = 0.16, \quad \kappa = 5, \quad \rho = 0.1. \end{aligned} \tag{8}$$

Table 1 gives the prices obtained using the AES for various choices of initial asset prices, as well as previously mentioned reference prices.

Number of Steps	$S_0 = 8$	$S_0 = 9$	$S_0 = 10$	$S_0 = 11$	$S_0 = 12$
$M = 3$	1.942	1.079	0.491	0.190	0.067
$M = 6$	1.971	1.101	0.510	0.204	0.076
$M = 12$	1.986	1.109	0.519	0.211	0.080
$M = 24$	1.993	1.113	0.523	0.214	0.082
$M = 60$	1.997	1.115	0.525	0.216	0.083
$M = 120$	1.999	1.116	0.526	0.216	0.083
Reference prices	1.999	1.108	0.521	0.215	0.083

Table 1. Prices of American put options under the Heston model using AES.

Figures 1 and 2 show high accuracy of the AES scheme with a small number of steps. When using standard polynomials as basis functions, a satisfactory price is obtained when using approximately $M = 20$ steps. However, when using Laguerre polynomials, the accuracy varies depending on whether the options are in-the-money, at-the-money, or out-of-the-money.

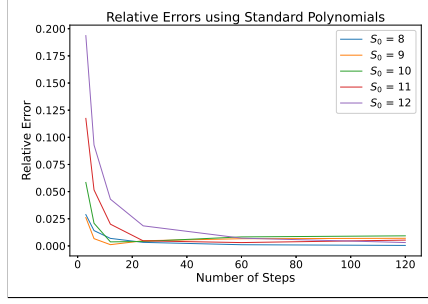


Figure 1. Relative Errors of American put option prices with parameters given in Table 2 calculated using AES and standard polynomials under the Heston Model.

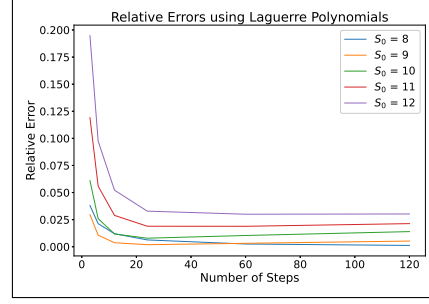


Figure 2. Relative Errors of American put option prices with parameters given in Eq. (8) calculated using AES and Laguerre polynomials under the Heston Model.

4. References

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\end{thebibliography}

```

Each entry takes the form:

```
\bibitem{key} Bibliography entry
```

where ‘key’ is the tag that is to be used as an argument for the `\cite{}` commands in the text of the article and ‘Bibliography entry’ is the material that is to appear in the list of references, suitably formatted. The commands

```

\usepackage[numbers,sort&compress]{natbib}
\bibpunct[, ]{[{}]}{,}{n}{,}{,}
\renewcommand\bibfont{\fontsize{10}{12}\selectfont}

```

need to be included in the preamble of your .tex file in order to generate the citations and bibliography as described above.

Instead of typing the bibliography by hand, you may prefer to create the list of references using a BIB_{TEX} database. In order to do this, the `tfs.bst` file needs to be in your working folder or an appropriate directory, and the lines

```

\bibliographystyle{tfs}
\bibliography{interacttfssample}

```

included where the list of references is to appear, where `tfs.bst` is the name of the BIB_{TEX} bibliography style file for Taylor & Francis’ Reference Style S and `interacttfssample.bib` is the bibliographic database included with the Interact-TFS L^AT_EX bundle (to be replaced with the name of your own .bib file). L^AT_EX/BIB_{TEX} will extract from your .bib file only those references that are cited in your .tex file and list them in the References section.

Please include a copy of your .bib file and/or the final generated .bbl file among your source files if your .tex file does not contain a reference list in a `thebibliography` environment.

5. Appendices

Any appendices should be placed after the list of references, beginning with the command `\appendix` followed by the command `\section` for each appendix title, e.g.

```

\appendix
\section{This is the title of the first appendix}
\section{This is the title of the second appendix}

```

produces:

Appendix A. This is the title of the first appendix

Appendix B. This is the title of the second appendix

Subsections, equations, figures, tables, etc. within appendices will then be automatically numbered as appropriate. Some theorem-like environments may need to have their counters reset manually (e.g. if they are not numbered within sections in the main text). You can achieve this by using `\numberwithin{remark}{section}` (for example) just after the `\appendix` command.

Note that if the `endfloat` package is used on a document containing any appendices, the `\processdelayedfloats` command must be included immediately before the `\appendix` command in order to ensure that the floats belonging to the main body of the text are numbered as such.

Appendix A. Troubleshooting

Authors may occasionally encounter problems with the preparation of a manuscript using L^AT_EX. The appropriate action to take will depend on the nature of the problem:

- (i) If the problem is with L^AT_EX itself, rather than with the actual macros, please consult an appropriate L^AT_EX 2_ε manual for initial advice. If the solution cannot be found, or if you suspect that the problem does lie with the macros, then please contact Taylor & Francis for assistance (latex.helpdesk@tandf.co.uk), clearly stating the title of the journal to which you are submitting.
- (ii) Problems with page make-up (e.g. occasional overlong lines of text; figures or tables appearing out of order): please do not try to fix these using ‘hard’ page make-up commands – the typesetter will deal with such problems. (You may, if you wish, draw attention to particular problems when submitting the final version of your manuscript.)
- (iii) If a required font is not available on your system, allow T_EX to substitute the font and specify which font is required in a covering letter accompanying your files.

Appendix B. Obtaining the template and class file

B.1. Via the Taylor & Francis website

This article template and the `interact` class file may be obtained via the ‘Instructions for Authors’ pages of selected Taylor & Francis journals.

Please note that the class file calls up the open-source L^AT_EX packages `booktabs.sty`, `epsfig.sty` and `rotating.sty`, which will, for convenience, unpack with the downloaded template and class file. The template calls for `natbib.sty` and `subfig.sty`, which are also supplied for convenience.

B.2. Via e-mail

This article template, the `interact` class file and the associated open-source L^AT_EX packages are also available via e-mail. Requests should be addressed to latex.helpdesk@tandf.co.uk, clearly stating for which journal you require the template and class file.